Julien Petit--Colboc

Engineering student in computer science – AI, VR and robotics
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Rouen, France
Portfolio – sliiz.com

Summary

4th-year engineering student in computer science (work-study) at CESI, apprentice at CAPFI since January 2026. Lead developer of Cerebro, an LLM-based matching system in production, where I worked on making its scores explainable and measurable. Previously a research intern at CESI LINEACT on teleoperation of robots in virtual reality. I am looking for an internship abroad from June 2027 (12 to 17 weeks) in AI, VR or robotics, ideally within a research team.

Online profiles

GitHub
SliiZ-code

LinkedIn
julien-petit-colboc

Languages

French
Native
●●●●●

English
Professional, TOEIC planned
●●●●○

German
Beginner
●●○○○

Skills

AI and LLMs
LLM pipelines, Prompt engineering, Structured outputs, OpenAI and Gemini APIs, Stable Diffusion, Whisper

Programming languages
Python, C#, C++, Dart, JavaScript

Back-end
FastAPI, .NET, REST API, SQLite, Pydantic

VR, 3D and electronics
Unity, Arduino

DevOps and systems
Docker, GitLab CI, Git Flow, Linux/WSL, VMware, Wireshark

Interests

Running (half marathon, March 2027), calisthenics, graphic design, travel

Experience

CAPFI — Paris
Work-study engineer (R&D, AI) — Jan 2026 – Present
Lead developer of Cerebro, an internal tool that matches consultants' profiles to client tenders using LLMs.

- Designed an AI pipeline that extracts, categorises and scores tender × profile pairs, with reliable asynchronous batch processing; deployed to production after several hundred tenders were processed.
- Calibrated the scoring prompt (temperature, required or preferred skills, three sub-scores) and added short justifications so users can verify each score.
- Launched an annotated reference set to measure false positives and false negatives.
- Built the daily report sent to about ten sales staff; secured the API and web interface with authentication and SSO. *Stack: Python, FastAPI, Pydantic, SQLite, Docker, GitLab CI, Gemini.*

CESI LINEACT — Rouen
Research intern — Apr 2025 – Jul 2025
VR application for teleoperating robots (Scrum; C#, .NET, Unity, REST API).

- Integrated the feed from a 360° camera mounted on the robot into an immersive VR view.
- Secured robot access with AES-256 encryption.
- Designed a real-time interface for editing waypoints.

Projects

Blink — 2024
Python pipeline that generates short videos from a simple topic: GPT-3.5 writes the script, Stable Diffusion XL creates the images, OpenAI TTS records the voiceover and Whisper generates the subtitles. Asynchronous API calls, automated editing with MoviePy.

Thrive — 2024
Cross-platform task manager (Android, iOS, Windows, macOS, Web) built with Flutter/Dart: Isar local database, categories, notifications.

Education

CESI École d'Ingénieurs — Engineering degree (master's level)
Computer science, work-study/apprenticeship — Rouen • Oct 2025 – Sep 2028

CESI — Integrated preparatory programme
Computer science — Rouen • Sep 2023 – Jul 2025

- Conway's Game of Life in C++ (pair project, main contributor): MVC architecture, SFML console and graphical modes, toroidal grid, obstacle cells.

Activities

BDE CESI Rouen – Head of Communication — Oct 2025 – Present
Managing the student association's communications: Instagram content, event posters, photo and video coverage.

CESI Robotics Club — 2023 – 2024
Developed remote control of a BB-8 replica: web joystick (HTML, JS) and a Flask server that drives the Arduino over serial link.